

Justine Cauvi

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Education

PhD in computer science (DI ENS, Paris, France)	2026-now
Title: Algorithmic aspects of ordered graphs Supervisor: Laurent Viennot	
Ecole Normale Supérieure de Lyon (ENS de Lyon, Lyon, France)	2022 - 2026
• "Projet Long de Recherche"	2025 - 2026
The "Projet Long de Recherche" is an academic year of the ENS de Lyon that consists of two research internships.	
• Master's Degree in Computer Science (with highest honour)	2023 - 2025
During my first year of master, I studied parallel and distributed algorithms, performance evaluation, optimization, approximation algorithms, data base and data mining as well as computational complexity, cryptography and security, quantum computer science, program analysis and German. During my second year of master, I studied distributed algorithms for networks, complex networks, data-aware algorithms for tensors, robust discrete optimization, machine learning for/with graphs, molecular programming and German.	
• Bachelor's Degree in Computer Science (with highest honour)	2022 - 2023
I studied theoretical and applied computer science and mathematics. Courses I've followed include algorithms and data-structures, graph theory, networking and Linux-based operating systems' architectures, probability, programming language theory, computability and formal logic. I also took English and German courses.	
"Classes préparatoires" (Lycée Masséna, Nice, France)	2020 - 2022
Two-year undergraduate intensive preparation courses for the entrance exams to top French engineering and research schools. I studied mathematics, physics, computer science and English.	
"Baccalauréat" (Lycée Audibert, Antibes, France)	2020
High School diploma in sciences, computer science option (with highest honour)	

Work experience

Research internship at University of Freiburg (Freiburg, Germany)	March 2026 - July 2026
Distributed Complexity of Locally Checkable Labeling Problems in Graphs of Unbounded Degree In the team of Fabian Kuhn, I worked on establishing a better understanding of the complexity landscape of Locally Finite Labelings, a generalization of Locally Checkable Labelings to unbounded degree graphs.	
Research internship at LIP6, Sorbonne Université (Paris, France)	October 2025 - February 2026
Routing and connectivity in temporal graphs Under the supervision of Thomas Bellitto and Bruno Escoffier, I worked on questions related to connectivity in temporal graphs. More precisely, I studied the problem that consists of constructing a temporal graph with minimum number of temporal edges that satisfies given reachability queries.	
Research internship at INRIA (Paris, France)	February - July 2025
Temporal graphs: labelings and parameters Under the supervision of Laurent Viennot, I worked on labeling schemes for temporal graphs. I also studied the problem of finding a restless temporal path parameterized by a recently introduced temporal parameter. In collaboration with Nils Morawietz, we studied realization problems for temporal graphs, focusing specifically on the realization of given path metrics.	

- Research internship at University of Konstanz (Konstanz, Germany)** May - July 2024
Landmark Hub Labeling for fast route planning
Under the supervision of Sabine Storandt, I worked on route planning algorithms in road networks, studying the Landmark Hub Labeling technique which extends the Hub Labeling concept.
- Research internship at INRIA (Sophia Antipolis, France)** June - July 2023
Lower bounds for the cutwidth minimization problem
In the team of David Coudert, I worked on graph theory, studying the cutwidth minimization problem, more specifically computing good lower bounds to improve the efficiency of a branch and bound algorithm.

Publications

- Designing Sparse Temporal Graphs Satisfying Connectivity Requirements** 2026
Thomas Bellitto, Jules Bouton Popper, Justine Cauvi, Bruno Escoffier, Raphaëlle Maistre
SAND 2026 - 5th Symposium on Algorithmic Foundations of Dynamic Networks (10.4230/LIPIcs.SAND.2026.15)
- Foremost, Fastest, Shortest: Temporal Graph Realization under Various Path Metrics** 2026
Justine Cauvi, Nils Morawietz, Laurent Viennot
STACS 2026 - 43rd International Symposium on Theoretical Aspects of Computer Science, Volume 364, pp. 24:1-24:19 (10.4230/LIPIcs.STACS.2026.24)
- Parameterized Restless Temporal Path** 2025
Justine Cauvi, Laurent Viennot
FCT 2025 - 25th International Symposium on Fundamentals of Computation Theory, Lecture Notes in Computer Science, vol 16106 (10.1007/978-3-032-04700-7_7)
- Exact and Approximate Hierarchical Hub Labeling** 2025
Justine Cauvi, Ruoying Li, Sabine Storandt
Journal of Graph Algorithms and Applications, 29(2), 103–126. (10.7155/jgaa.v29i2.3041)
- Landmark Hub Labeling: Improved Bounds and Faster Query Answering (Best Paper Award)** 2024
Justine Cauvi, Ruoying Li, Sabine Storandt
ATMOS 2024 - 24th Symposium on Algorithmic Approaches for Transportation Modelling, Optimization, and Systems, Volume 123, pp. 1:1-1:17. (10.4230/OASlcs.ATMOS.2024.1)

Talks

- Foremost, Fastest, Shortest: Temporal Graph Realization under Various Path Metrics** 2026
Meeting of ANR TEMPOGRAL
Chamarande, France
- Foremost, Fastest, Shortest: Temporal Graph Realization under Various Path Metrics** 2026
International Symposium on Theoretical Aspects of Computer Science (STACS)
Grenoble, France
- Temporal graphs: parameters, labelings and realizations** 2025
Operation Research team seminar, LIP6, Sorbonne Université
Paris, France
- Parameterized Restless Temporal Path** 2025
Modern Optimization Methods and Models in Algorithm Design school-workshop (MoMA)
Aussois, France
- Parameterized Restless Temporal Path** 2025
Journées Graphes et Algorithmes (JGA)
Paris, France

Parameterized Restless Temporal Path	2025
International Symposium on Fundamentals of Computation Theory (FCT) Wrocław, Poland	
Landmark Hub Labeling: Improved Bounds and Faster Query Answering	2024
Seminar at IMB and LaBRI Bordeaux, France	

Attended events

Meeting of ANR TEMPOGRAL	August 2026
Chamarande, France	
SAND 2026	July 2026
Le Havre, France	
STACS 2026	March 2026
Grenoble, France	
School-workshop on Modern Optimization Methods and Models in Algorithm Design (MoMA)	November 2025
Aussois, France	
Journées Graphes et Algorithmes (JGA)	November 2025
Paris, France	
ALGO2025	September 2025
Warsaw, Poland	
FCT 2025	September 2025
Wrocław, Poland	
Meeting of ANR TEMPOGRAL	May 2025
Île d'Oléron, France	
ALGO2024	September 2024
Egham, United Kingdom	

Teaching experiences

Algorithms and Complexity Seminar	Summer Term 2026
University of Freiburg, Germany	
Supervision of two groups of students presenting a research paper on MPC algorithms	

Projects

Integrated project: a pedagogical proof assistant	2023 - 2024
With a group of five, we built a prototype of proof assistant for the integrated project computer science course at ENS de Lyon (using lambda calculus and web technologies to make a proof assistant more accessible).	
Performance Evaluation project	2023
I studied and compared genetic algorithms applied on a road network congestion problem.	
“Classes préparatoires” project	2021 - 2022
I studied exact and approximation algorithms for the traveling salesman problem.	

Skills

Languages: French (native), English (C1), German (B2)
Programming languages: Python, C, C++, OCaml, LaTeX

Certificates

- Cambridge C1 Advanced
- Driving license